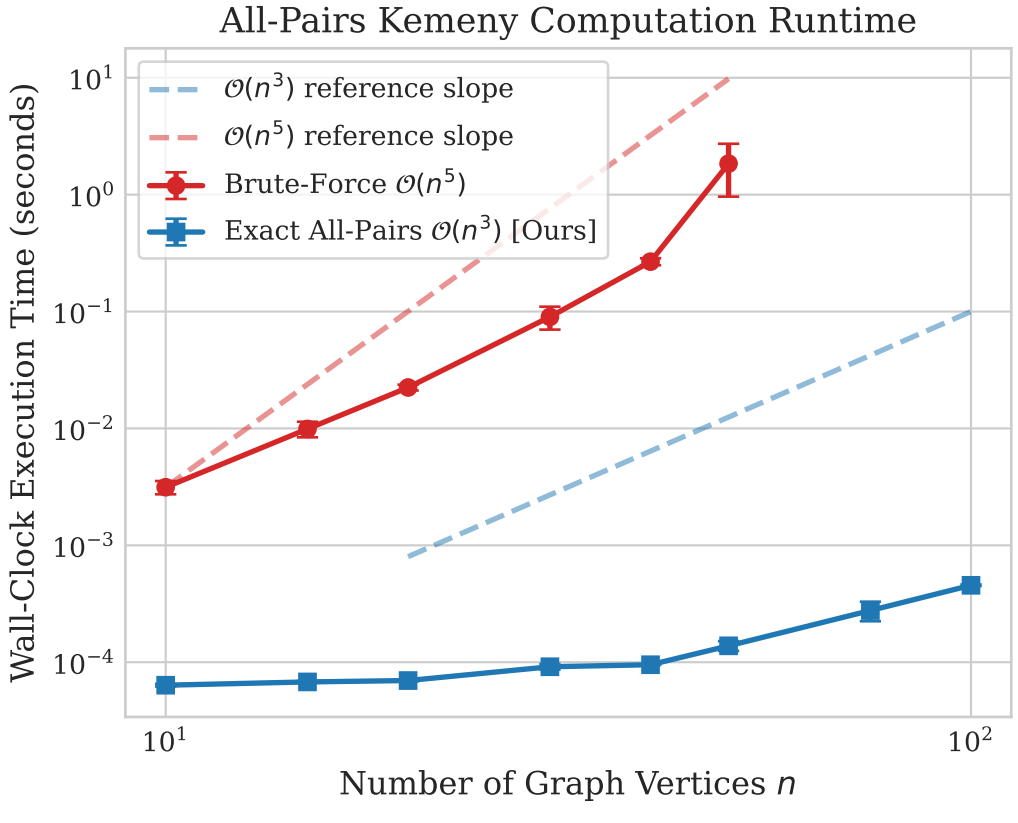
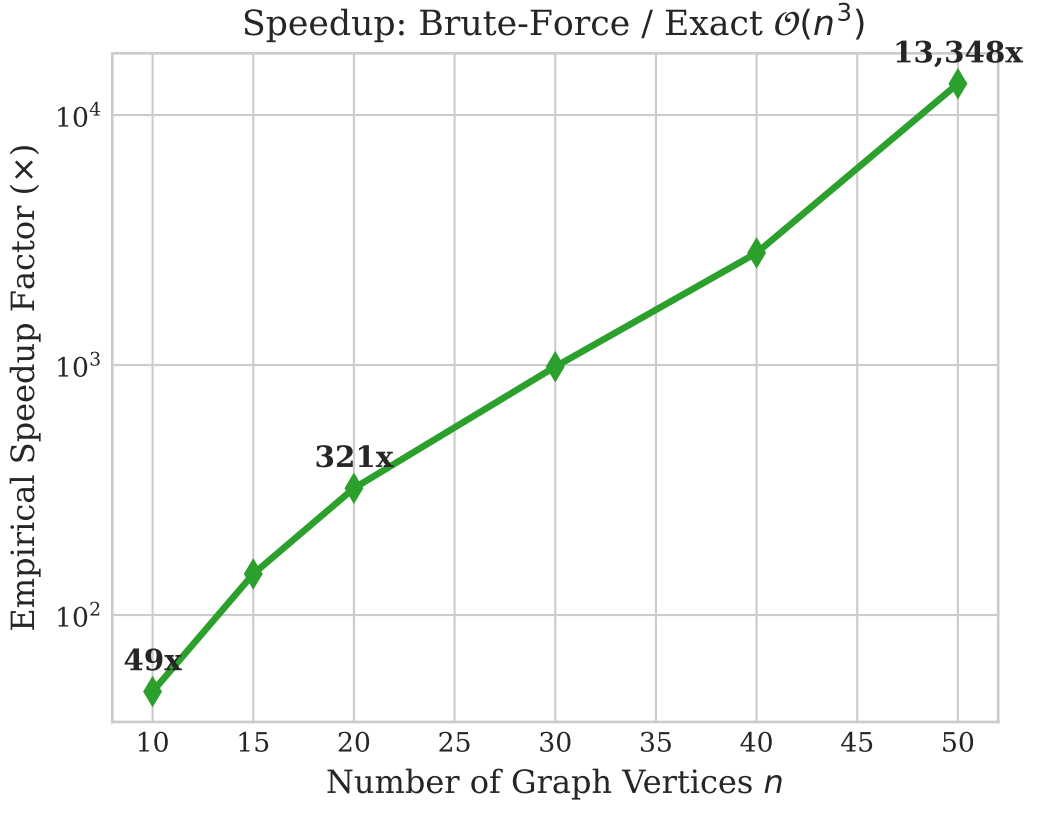

All-Pairs Kemeny Computation Runtime
$\mathcal{O}(n^3)$ reference slope
$\mathcal{O}(n^5)$ reference slope
Brute-Force $\mathcal{O}(n^5)$
Exact All-Pairs $\mathcal{O}(n^3)$ [Ours]
Wall-Clock Execution Time (seconds)
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{1}
10^{2}
Number of Graph Vertices n


Speedup: Brute-Force / Exact $\mathcal{O}(n^3)$
Empirical Speedup Factor (×)
10^{4}
10^{3}
10^{2}
49x
321x
13,348x
10
15
20
25
30
35
40
45
50
Number of Graph Vertices n